6.1. Clarity

A function describes a computational process. It's important that the process that the function describes is represented clearly and accurately. You'll begin with function composition and then look at a specific problem and how to describe it with functions. You'll start with an optimization problem: how can Agtron maximize the profit from his online store when sales figures vary depending on sales prices?

By tracking purchases on his shop website, Agtron has found that when he sets the price for the PhotomakerExtreme (a camera) at $200, he sells (on average) 50 per day. For every $20 that he reduces the price, he sells an additional 10 units per day. Each PhotomakerExtreme costs Agtron $100 from the wholesaler, and it costs $140 per day to run the website.

To solve this problem by composing functions, you'll define functions by describing what they do—using *other* functions.

6.1.1. Functions are descriptions

When you read the body of a function, the intention and meaning should be clear. Suppose your esteemed colleague Mr. Rob Tuse has previously written a function to evaluate the profit for a given `price`:

```
profit = (50+(20/10)*(200-price))*price-(140+(100*(50+((20/10)*
(200-price)))))
```

Aside from clues that the words `price` and `profit` afford, can you tell what the function means? Can Rob? Not likely. How do you define `profit` to clearly indicate what it is? Starting by thinking about the relationship. The `profit` for a sale price is the `revenue` at that price minus the `cost` at that price:

```
profit = (salePrice) ->
  (revenue salePrice) – (cost salePrice)
```